Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/05/2005
PI Name:	Sack, Fred Ph.D.		
Project Title:	Development of Gravity Sensitive Plant Cells (Ceratodon) in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sack.1@osu.edu	**Fax:**	FY 614-292-6345
PI Organization Type:	UNIVERSITY	**Phone:**	614-292-0896
Organization Name:	Ohio State University		
PI Address 1:	318 W. 12th Ave.		
PI Address 2:	500 Aronoff Building		
PI Web Page:			
City:	Columbus	**State:**	OH
Zip Code:	42310	**Congressional District:**	15
Comments:	Deceased August 2015.		
Project Type:	Ground	**Solicitation / Funding Source:**	93-OLMSA-01
Start Date:	07/15/1999	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle		
Flight Assignment:	STS-107		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG10-0263		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Little is known about how gravity regulates the distribution of high-density organelles or the direction of plant cell growth. Dark-grown cells of the moss Ceratodon grow away from the gravity vector. These cells probably use amyloplasts that sediment to sense the g- vector. Only a subset of amyloplasts sediment along the length of the cell. If gravity were the sole determinant of plastid distribution and cell orientation, then both would be randomly distributed in microgravity. Instead, both were found to be non-random in microgravity (STS-87, SPM-A). This work was extended by flying the BRIC-14 experiment on STS-107. Our material was chemically fixed in space and our hardware was recovered on the ground after the Shuttle broke up. We were able to recover cultures from the hardware and photograph cells. The fixative imparted enough integrity to the moss that we were able to spiral growth and cellular preservation. Most plants flown in space show a random distribution of organs that normally would be oriented by gravity on earth. However, moss cultures showed a spiral pattern suggesting the existence of an underlying default growth pattern that is normally		

	masked by the gravity response.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work addresses fundamental questions about how the internal organization and the growth pattern of specialized cells are affected by gravity.
Task Progress:	Our spaceflight results with spirals recovered from STS 107 were published in the peer-reviewed plant journal Planta in May, 2005. In addition intracellular data about the role of the position of organelles of high mass are being evaluated from space grown and control cells.
Bibliography Type:	Description: (Last Updated: 05/05/2008)
Articles in Peer-reviewed Journals	Kern VD, Schwuchow JM, Reed DW, Nadeau JA, Lucas J, Skripnikov A, Sack FD. "Gravitropic moss cells default to spiral growth on the clinostat and in microgravity during spaceflight." Planta. 2005 Apr;221(1):149-57. Epub 2005 Jan 20. PMID: 15660206 , Apr-2005